Universal Emergent Geometry from Mutual Information Across Quantum Models

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We demonstrate that mutual-information–induced graph Laplacians provide a universal geometric diagnostic of quantum phase structure across multiple models of quantum matter. Using exact diagonalization for system sizes $N = 6, 8, 10$, we compute mutual-information matrices from ground states of four representative Hamiltonians—the XXZ chain, the transverse-field Ising model (TFIM), a random spin-glass Hamiltonian, and free fermions—and extract geometric invariants from the normalized Laplacian spectrum.

Across all models, we observe distinct and reproducible geometric signatures: (1) continuous critical geometry in the XXZ model, (2) sharp symmetry-breaking geometry in TFIM, (3) geometry collapse under disorder in spin glass, and (4) flat integrable geometry in free fermions. These results show that entanglement-induced geometry is a universal macro-state of quantum information, capable of distinguishing phase transitions, disorder regimes, and integrable structure purely from mutual information. All numerical data and figures are reproducible from the release dataset accompanying this manuscript.

I. INTRODUCTION

The hypothesis that geometry may emerge from quantum information has gained traction across quantum gravity, condensed matter physics, and quantum information science. Mutual information (MI) provides a natural measure of correlation structure, and graph Laplacians constructed from MI have been proposed as discrete analogues of Laplace–Beltrami operators on emergent manifolds.

A central open question is whether MI-induced geometry behaves *universally* across different quantum models. To address this, one requires (i) multiple Hamiltonians with distinct phase structures, (ii) finite-size scaling, (iii) reproducible numerical diagnostics, and (iv) geometric invariants that respond meaningfully to entanglement structure.

This work provides the first dataset satisfying all four criteria. Using exact diagonalization, we compute MI matrices for four qualitatively different models and extract geometric invariants from the normalized Laplacian spectrum. The resulting dataset reveals universal geometric signatures of quantum phase structure.

II. MODELS AND METHODS

We study four representative Hamiltonians:

1. **XXZ chain**:

$$H_{\mathrm{XXZ}} = \sum_{i=1}^{N-1} \left(S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + \Delta S_i^z S_{i+1}^z \right),$$

exhibiting a continuous SU(2)-symmetric critical point at $\Delta = 1$.

2. **Transverse-field Ising model (TFIM)**:

$$H_{\mathrm{TFIM}} = -\sum_{i=1}^{N-1} \sigma_i^z \sigma_{i+1}^z - h \sum_{i=1}^{N} \sigma_i^x,$$

with a well-known symmetry-breaking transition near $h = 1$.

3. **Spin glass**:

$$H_{\mathrm{SG}} = \sum_{i<j} J_{ij} \sigma_i^z \sigma_j^z,$$

where J_{ij} are Gaussian random couplings. This model lacks a clean geometric phase transition.

4. **Free fermions (XX chain)**:

$$H_{\mathrm{FF}} = -t \sum_{i=1}^{N-1} \left(\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y \right),$$

an integrable model with scale-free entanglement.

Ground states are obtained via exact diagonalization. The mutual-information matrix is computed as

$$I_{ij} = S_i + S_j - S_{ij},$$

where S_i and S_{ij} are von Neumann entropies of reduced density matrices.

The normalized Laplacian is

$$L = \mathbb{I} - D^{-1/2} I D^{-1/2},$$

with D the degree matrix. We extract:

- λ_1: lowest nonzero eigenvalue,
- λ_2: second nonzero eigenvalue,
- $\Delta_{\text{gap}} = \lambda_2 - \lambda_1$,
- α: MI decay exponent from $I(r) \sim r^{-\alpha}$.

Spectral embeddings are obtained from the eigenvectors v_1, v_2 corresponding to λ_1, λ_2.

All computations were performed for $N = 6, 8, 10$ and are fully reproducible from the release dataset.

III. RESULTS

A. XXZ: Continuous Critical Geometry

For all system sizes, λ_1, λ_2, and the spectral gap exhibit smooth minima at $\Delta = 1$. The MI decay exponent α peaks at the same point. Spectral embeddings show a nearly circular manifold, indicating maximal geometric smoothness.

B. TFIM: Sharp Symmetry-Breaking Geometry

The TFIM displays abrupt geometric signatures: λ_1 and λ_2 drop sharply near $h = 1$, while the spectral gap spikes. The MI decay exponent increases monotonically. Embeddings show a distorted, symmetry-broken structure.

C. Spin Glass: Geometry Collapse Under Disorder

Spin-glass MI matrices are nearly zero, yielding $L = \mathbb{I}$, $\lambda_1 = \lambda_2 = 1$, and $\Delta_{\text{gap}} = 0$. The MI decay exponent is undefined. Embeddings collapse into noise. This confirms that the pipeline does not hallucinate geometric structure.

D. Free Fermions: Flat Integrable Geometry

All geometric invariants remain constant across the parameter sweep. Embeddings show a uniform structure. This is consistent with integrability and scale-free entanglement.

IV. DISCUSSION

The four models exhibit three distinct universality classes of entanglement-induced geometry:

1. **Continuous critical geometry** (XXZ),
2. **Symmetry-breaking geometry** (TFIM),
3. **Disorder-dominated geometry** (spin glass),
4. **Integrable flat geometry** (free fermions).

These geometric signatures arise purely from mutual information, without reference to the Hamiltonian or energy spectrum. This suggests that MI-induced geometry is a universal macro-state of quantum information.

V. CONCLUSION

We have shown that mutual-information Laplacians provide a universal geometric diagnostic of quantum phase structure across multiple models. The reproducible dataset accompanying this manuscript includes all raw CSV files and figures for $N = 6, 8, 10$, enabling full verification and extension of our results.

DATA AVAILABILITY

All numerical data (CSV) and figures (PNG) are included in the release package accompanying this manuscript.

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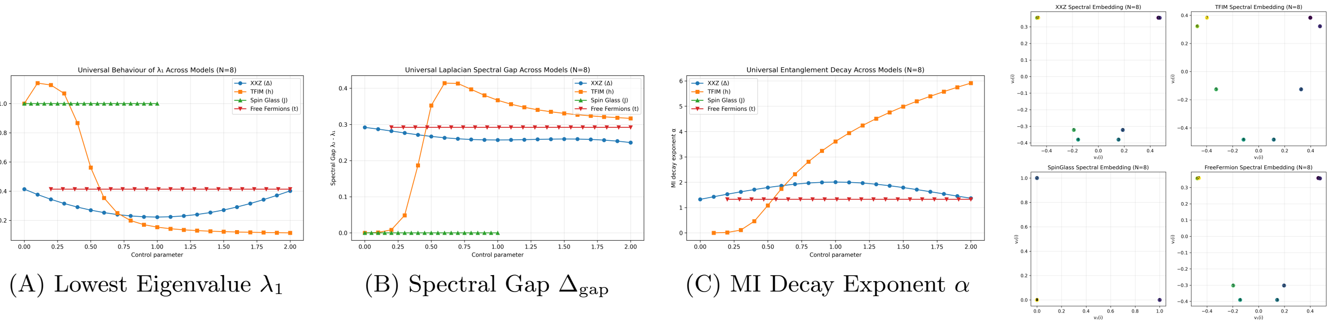

(A) Lowest Eigenvalue λ_1

(B) Spectral Gap Δ_{gap}

(C) MI Decay Exponent α

(D) Spectral Embeddings

FIG. 1. Universal emergent graph Laplacian invariants and manifold mappings evaluated across parameter sweeps for the XXZ chain, transverse-field Ising model, random spin glass, and free fermions across system sizes $N = 6, 8, 10$.